

Executive Overview

Peak Demand in MW

122K

Peak Demand

Average Demand in MW

92.75K

Avg Demand

EIA MAPE %

3.02

EIA MAPE %

Load Factor KPI

76.2%

Load Factor

Date Filter

Date

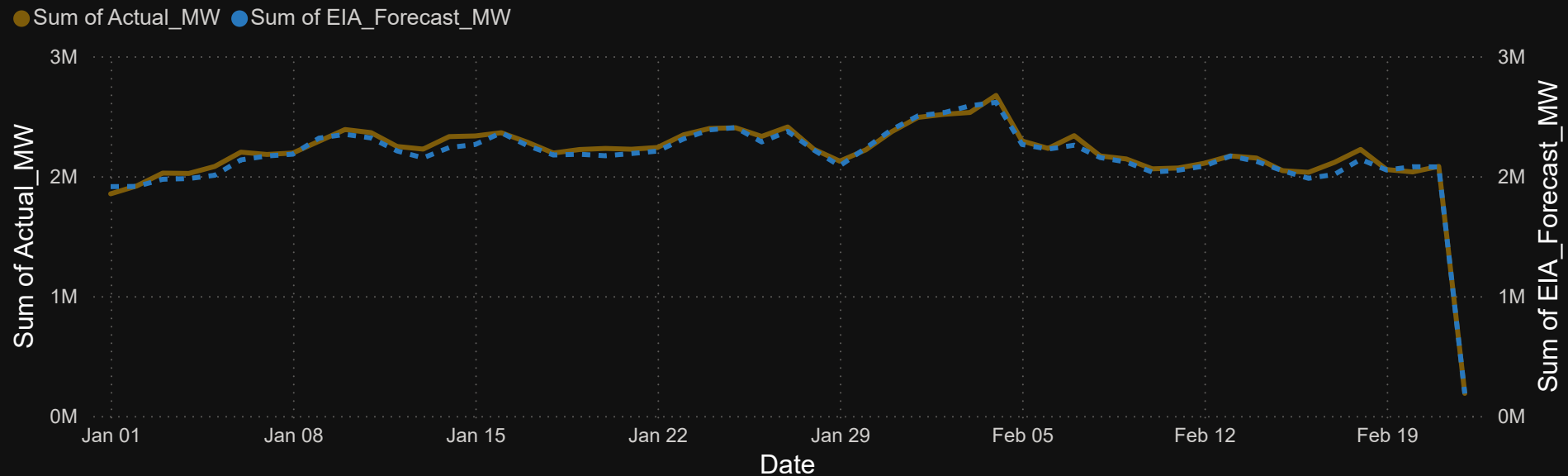
1/1/2023



2/22/2023



Actual_MW and EIA_Forecast_MW by Date

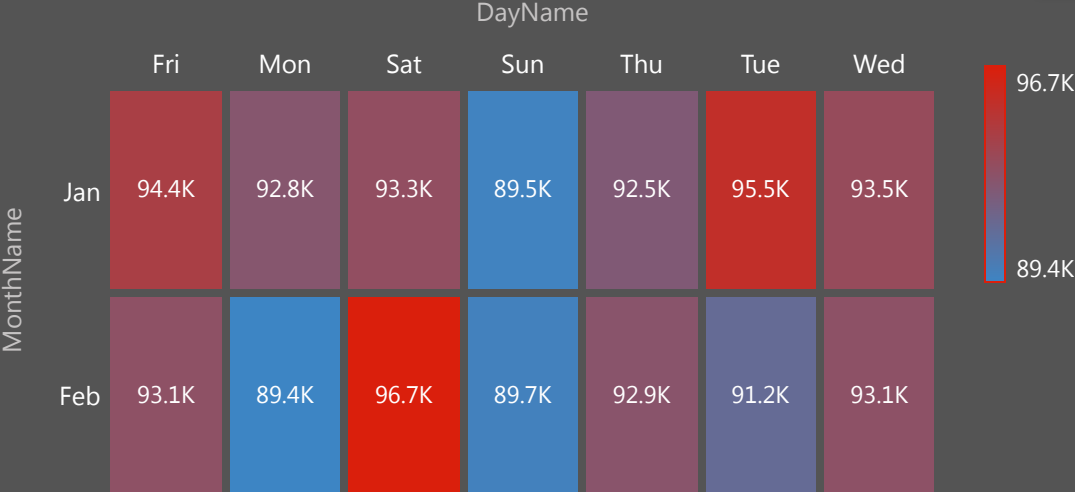


Ouaomar
Marouane

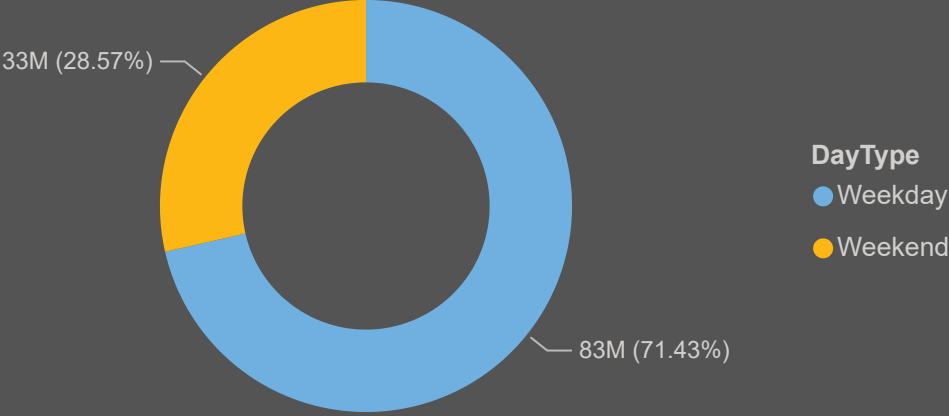
"PJM Energy
Demand Analysis"

Demand Patterns

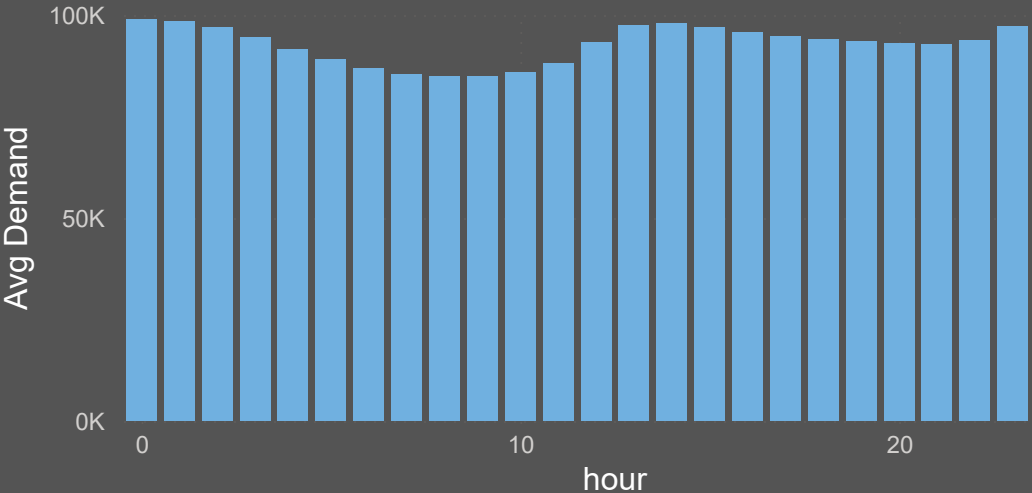
Average Demand by Month and Day



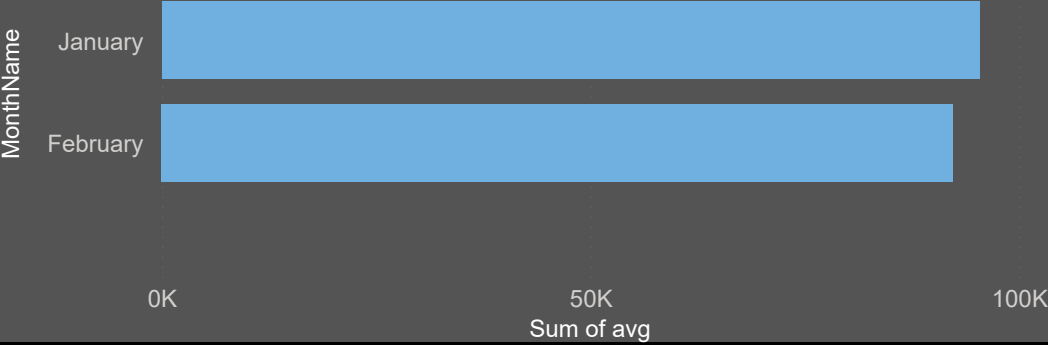
Weekend VS Weekday demand in MW



Average Demand by hour



Average Demand (Mw) by Month



Model Performance

Model Metrics				
Model	Sum of RMSE	Sum of MAPE (%)	Sum of MAE	Sum of Train time (s)
EIA Forecast (DF)	3,435.60	3.02	2,808.40	
Prophet	16,025.80	15.25	14,403.90	0.30
SARIMA	23,451.60	21.38	18,886.90	9.38
XGBoost	3,986.80	3.11	2,896.80	0.82

